Experiment 7: Android Calculator Application

AIM:

Todevelop an Android application that performs basic arithmetic operations such as addition, subtraction, multiplication, and division using Android UI components.

PROCEDURE:

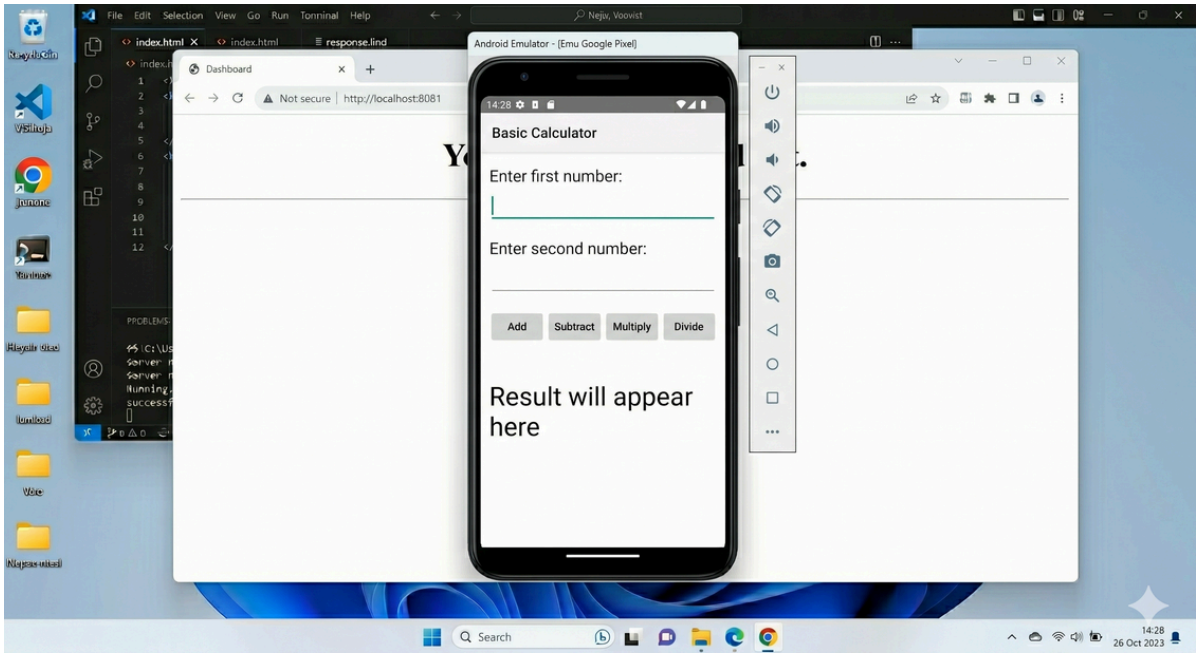
1. Open Android Studio and create a new project.
2. Choose Empty Views Activity template.
3. Design UI using EditText, Buttons, and TextView in XML.
4. Link UI components in MainActivity.java.
5. Retrieve user inputs and convert to numbers.
6. Set OnClickListeners for operation buttons.
7. Perform arithmetic operations.
8. Display result in TextView.
9. Handle invalid inputs and division by zero.
10. Run the application using emulator.

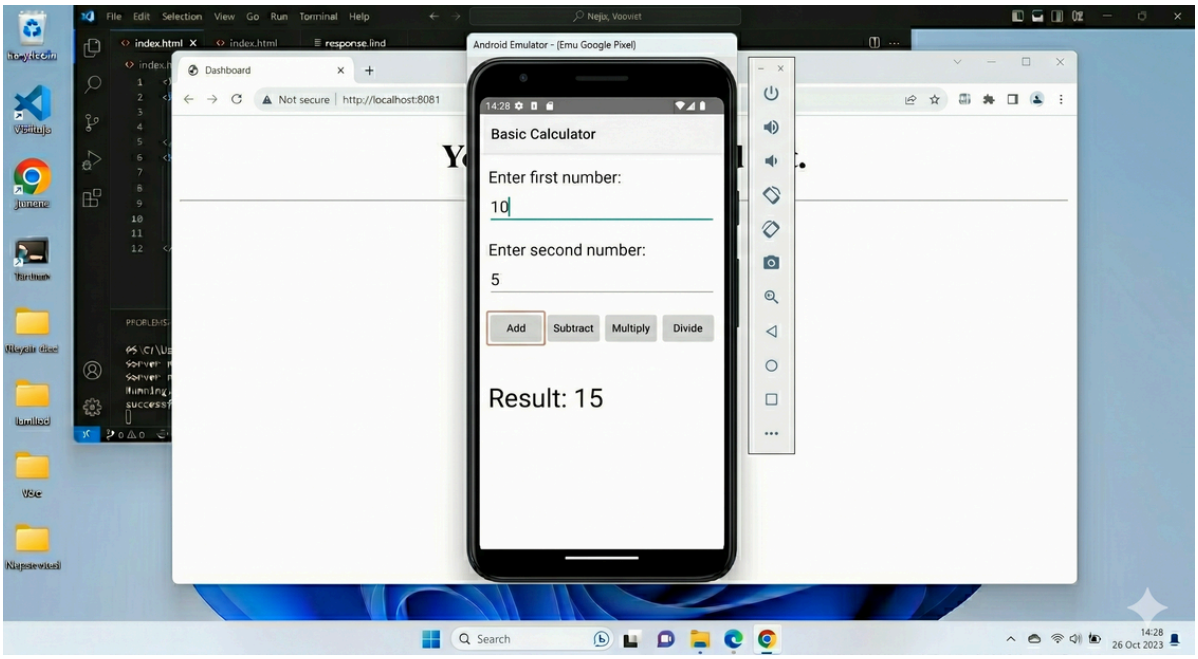
CODE:

XMLLayout:
EditText (2), Buttons (Add, Subtract, Multiply, Divide, Clear), TextView for result.
Java Logic:

```
double num1 = Double.parseDouble(input1);
double num2 = Double.parseDouble(input2);
switch(operator){
case '+': result = num1 + num2; break;
case '-': result = num1 - num2; break;
case '*': result = num1 * num2; break;
case '/': if(num2==0) show error;
}
textView.setText("Result: " + result);
```

OUTPUT:





RESULT:

An Android calculator application is successfully developed to perform basic arithmetic operations with proper input validation.